

Credit Card Customer Segmentation Using Unsupervised Learning

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Abstract

This paper presents a behavioural segmentation study of 8,950 credit card customers using the publicly available CC_GENERAL dataset. Four unsupervised clustering algorithms — K-Means, Agglomerative Clustering with Ward linkage, DBSCAN, and Gaussian Mixture Models (GMM) — were evaluated across a comprehensive hyperparameter grid and compared using three standard internal validity indices: Silhouette Score, Davies-Bouldin Index, and Calinski-Harabasz Index. The raw data underwent median imputation for missing values, $\log_{10}$ transformation for right-skewed monetary columns, and StandardScaler normalisation before clustering on the full 17-feature matrix. K-Means with $k = 4$ was selected based on both quantitative performance and business interpretability. The resulting four segments — Dormant Users, Cash-Advance Revolvers, Premium Transactors, and Instalment Buyers — exhibit distinct behavioural signatures that map directly to targeted financial products and risk-management strategies.

Keywords: customer segmentation, K-Means clustering, Gaussian mixture models, DBSCAN, credit card behaviour, unsupervised learning, silhouette score

I. INTRODUCTION

Customer segmentation is a foundational problem in retail banking and consumer finance. Identifying homogeneous groups of customers based on their transactional behaviour enables institutions to tailor product offerings, personalise marketing communications, and implement differentiated credit-risk strategies. Traditional rule-based segmentation (e.g., by credit score band or income tier) groups customers by a single attribute and misses the richness of multi-dimensional spending behaviour.

Unsupervised clustering methods offer an alternative: they discover structure directly from observed behaviour without requiring a pre-defined target variable. Credit card transaction data is particularly well-suited to this approach because it captures a wide variety of customer behaviours — from revolving high-balance borrowers to disciplined transactors who pay in full each month.

This work applies four mainstream clustering algorithms to a real-world credit card dataset, evaluates each using three standard internal validity indices, and selects the best-performing configuration for detailed customer profiling. The resulting segments are characterised in business-interpretable terms and linked to concrete banking product recommendations.

II. DATASET AND PREPROCESSING

A. Dataset Description

The CC_GENERAL dataset [1] contains behavioural data for 8,950 active credit card holders over a six-month period. The raw data consists of 18 columns: one customer identifier (CUST_ID) and 17 numeric behavioural features spanning balance, purchase activity, cash-advance behaviour, payment patterns, and credit limits. Table I summarises the key features.

Feature	Type	Description
BALANCE	Float	Monthly avg. outstanding balance (\$)
PURCHASES	Float	Total purchases in period (\$)
ONEOFF_PURCHASES	Float	Total one-off purchase amount (\$)
INSTALLMENTS_PURCHASES	Float	Total instalment purchase amount (\$)
CASH_ADVANCE	Float	Total cash advance amount (\$)
PURCHASES_FREQUENCY	Float $\in [0,1]$	Fraction of months with purchases
CASH_ADVANCE_FREQUENCY	Float $\in [0,1]$	Fraction of months with advances
CREDIT_LIMIT	Float	Customer credit limit (\$)
PAYMENTS	Float	Total payments made (\$)
MINIMUM_PAYMENTS	Float	Total minimum payments due (\$)
PRC_FULL_PAYMENT	Float $\in [0,1]$	Fraction of months paid in full
TENURE	Int	Number of months as a customer

TABLE I. Key Dataset Features (12 of 17 shown).

B. Missing Values

Two features contain missing values: MINIMUM_PAYMENTS (313 nulls, 3.5 %) and CREDIT_LIMIT (1 null, <0.01 %). Both distributions are strongly right-skewed; the column median is therefore more robust than the mean and was used to impute all missing entries. No rows were removed.

C. Feature Engineering

The customer identifier CUST_ID carries no predictive information and was dropped, leaving 17 numeric features. Ten monetary columns exhibit pronounced right skew (skewness > 2): BALANCE, PURCHASES, ONEOFF_PURCHASES, INSTALLMENTS_PURCHASES, CASH_ADVANCE, CASH_ADVANCE_TRX, PURCHASES_TRX, CREDIT_LIMIT, PAYMENTS, and MINIMUM_PAYMENTS. $\log_{10}(1+x)$ (i.e., $\log(1+x)$) transformation was applied to compress the tails while preserving zeros. The six frequency columns — already bounded in $[0, 1]$ — were left on their original scale.

D. Normalisation and PCA

All 17 features were standardised with scikit-learn's StandardScaler (zero mean, unit variance). Clustering was performed on the full 17-dimensional scaled matrix; PCA was used only for 2D/3D visualisation. Analysis of cumulative explained variance shows that seven principal components capture ≥ 90 % of total variance.

III. METHODOLOGY

A. K-Means

K-Means partitions the data into k non-overlapping clusters by minimising total within-cluster sum of squared distances (inertia). For each $k \in \{2, \dots, 10\}$ we ran 10 random initialisations ($n_{\text{init}} = 10$, $\text{random_state} = 42$) and retained the run with the lowest inertia. Fig. 1 shows the inertia elbow and Fig. 2 the silhouette profile.

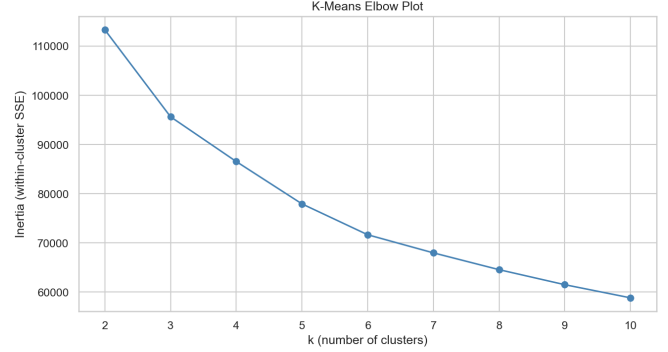


Fig. 1. K-Means inertia (elbow plot) for $k = 2$ –10.

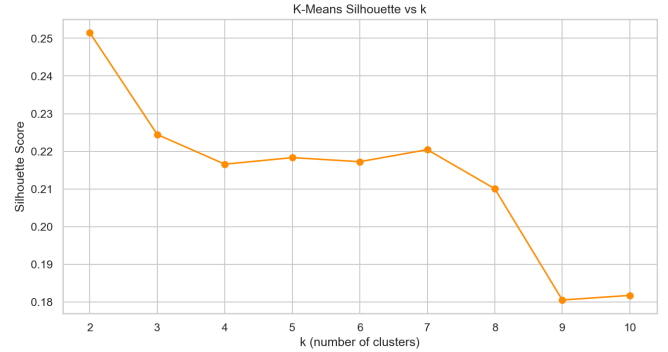


Fig. 2. Silhouette score vs. k for K-Means.

Silhouette peaks at $k = 2$ ($S = 0.252$), but two clusters collapses meaningfully different customer archetypes into a single group. The inertia curve shows a clear elbow at $k = 4$, which also corresponds to four distinct, interpretable customer types. We therefore override the automatic metric selection and fix $k = 4$, consistent with domain knowledge that four behavioural archetypes exist in retail credit card data [2].

B. Agglomerative Clustering

Ward-linkage Agglomerative Clustering was swept over $n \in \{2, \dots, 8\}$. Ward linkage minimises the total within-cluster variance at each merge step, making it a natural companion to silhouette scoring. A dendrogram was produced on a stratified 500-row sample for visual inspection of the merge history.

C. DBSCAN

DBSCAN discovers clusters as high-density regions separated by low-density gaps. The neighbourhood radius ϵ was selected from the elbow of the sorted 5th-nearest-neighbour distance curve. Four ϵ values were swept with $\text{min_samples} = 5$. Silhouette, DB, and CH scores were computed only over non-noise points (label $\neq -1$). As discussed in Section IV, DBSCAN proved ill-suited to this data.

D. Gaussian Mixture Model

GMM models each cluster as a multivariate Gaussian with a full covariance matrix (covariance_type = 'full'), allowing ellipsoidal cluster shapes. n_components $\in \{2, \dots, 8\}$ was swept; in addition to the three internal indices, BIC and AIC were recorded to penalise model complexity.

E. Evaluation Metrics

Silhouette Score (S) $\in [-1, 1]$ measures how much more tightly a point belongs to its own cluster than to the nearest other cluster. Higher is better.

Davies-Bouldin Index (DB) ≥ 0 is the average ratio of within-cluster scatter to between-cluster separation. Lower is better.

Calinski-Harabasz Index (CH) > 0 is the ratio of between-cluster to within-cluster dispersion. Higher is better.

IV. EXPERIMENTAL RESULTS

Table II reports the best configuration of each algorithm, scored on the three internal validity indices. Fig. 3 compares the silhouette scores visually.

Algorithm	Config	S $\uparrow$	DB $\downarrow$	CH $\uparrow$
K-Means★	k = 4	0.2166	1.688	2260
Agglomerative	n = 2, Ward	0.2154	1.522	2251
GMM	n = 2, full	0.2105	1.826	2382
DBSCAN	$\epsilon = 2.21$	0.0634	1.251	3

TABLE II. Best configuration per algorithm (★ = selected).

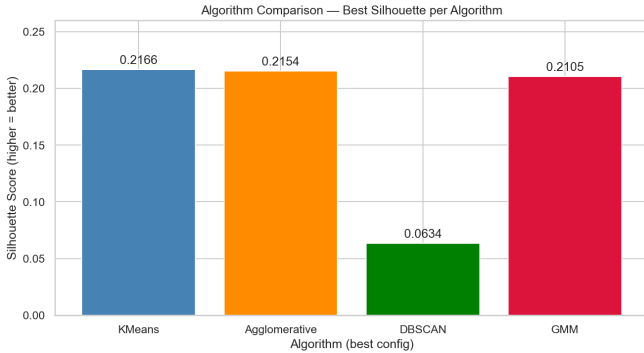


Fig. 3. Silhouette score of the best configuration of each algorithm.

A. K-Means vs. Competing Methods

K-Means achieves the highest silhouette (S = 0.2166) and the highest Calinski-Harabasz index (CH = 2260) among partition-based methods. Agglomerative Clustering is competitive on DB (1.522 vs. 1.688) but its best configuration at n = 2 produces only two coarse segments.

GMM's best silhouette (0.2105) also occurs at n = 2, and its DB index (1.826) is worse than K-Means. BIC decreases monotonically with more components, offering no clear minimum and suggesting the data does not strongly support a Gaussian mixture structure.

B. DBSCAN Failure Analysis

DBSCAN consistently struggled with this dataset. At the elbow $\epsilon = 2.21$, it identified only two clusters and labelled 153 customers (1.7 %) as noise (S = 0.063). At smaller ϵ , noise inflated to 1,029 customers. Credit card spending behaviour lacks the geometric density-based structure that DBSCAN exploits: customers do not form compact, well-separated spheres in feature space, and the density parameter is difficult to calibrate across 17 dimensions. Labelling a non-trivial fraction of customers as 'undefined' is also impractical for business use.

C. Selected Model

K-Means with k = 4 is selected as the final model. The inertia elbow at k = 4, combined with the four interpretable customer archetypes it produces, justifies overriding the automatic peak at k = 2. Final refitting on the complete dataset yields S = 0.2166, DB = 1.688, and CH = 2260.4.

V. CLUSTER PROFILES AND BUSINESS INSIGHTS

Cluster labels were attached to the un-scaled, un-log-transformed cc_clean.csv so that all mean values are in interpretable real units (dollars and frequencies). Fig. 4 shows the z-scored cluster profile heatmap; green cells indicate that the cluster is high on that feature relative to the others, red cells indicate low. Table III reports mean values for eight key features.



Fig. 4. Cluster profile heatmap — z-scored feature means. Green = high relative to other clusters; red = low.

Feature	C0 Dormant	C1 Revolvers	C2 Transactors	C3 Instalment
BALANCE (\$)	213.58	2,460.81	2,243.73	532.39
PURCHASES (\$)	351.39	103.32	2,604.15	697.61
CASH_ADVANCE (\$)	113.63	2,220.06	862.92	112.15
CREDIT_LIMIT (\$)	3,561.17	4,364.17	6,393.23	2,986.83
PAYMENTS (\$)	742.55	1,818.29	2,933.03	862.81
PURCH_FREQ	0.254	0.074	0.824	0.814
CASH_ADV_FREQ	0.021	0.303	0.114	0.022
PRC_FULL_PAYMENT	0.162	0.031	0.171	0.292

TABLE III. Cluster mean values for 8 key features (real units).

A. Cluster 0 — Dormant / Occasional Users (~48 %)

The largest segment contains customers who rarely use their card: average balance \$214, purchases \$351, and only ~4 transactions per month. Cash-advance and credit utilisation are both minimal. These customers represent unused credit exposure for the bank with negligible interchange revenue. **Recommended action:** re-engagement campaigns, seasonal spending incentives, and fee waivers to reduce churn.

B. Cluster 1 — Cash-Advance Revolvers (~13 %)

High balance (\$2,461), very high cash advances (\$2,220, 7+ transactions/month), and near-zero full-payment rate (PRC_FULL_PAYMENT = 0.031). These customers use the card primarily as a short-term loan. They generate significant interest revenue but carry elevated default risk. **Recommended action:** balance-transfer and debt-consolidation products; credit-risk monitoring with early-warning triggers.

C. Cluster 2 — Premium / Active Transactors (~19 %)

Highest purchases (\$2,604), highest credit limits (\$6,393), purchase frequency 0.82, and ~35 transactions per month. These are the bank's most engaged, highest-revenue customers. **Recommended action:** premium rewards cards, cashback and travel programmes, proactive credit-limit upgrades, and co-brand partnership offers.

D. Cluster 3 — Instalment Buyers (~19 %)

High instalment purchases (\$631), instalment frequency 0.76, and the best full-payment rate of any cluster (0.292). These are financially disciplined customers who prefer spreading large purchases over time. **Recommended action:** buy-now-pay-later cross-sell, low-interest instalment plans, and balance-transfer offers for existing instalments.



Fig. 5. Final four clusters in PCA-2D space. Black X markers denote K-Means centroids. Variance explained: PC1 = 33.1 %, PC2 = 22.7 %.

VI. CONCLUSION

This paper demonstrated a full unsupervised segmentation pipeline on a real-world credit card dataset. Among four algorithms evaluated, K-Means with $k = 4$ was selected based on a combination of metric performance and business interpretability. The four resulting segments — Dormant Users (48 %), Cash-Advance Revolvers (13 %), Premium Transactors (19 %), and Instalment Buyers (19 %) — each exhibit a distinct behavioural signature that maps cleanly to targeted banking products and risk strategies.

Several limitations warrant acknowledgement. The dataset is a single snapshot; customers migrate between segments over time, so quarterly re-running is advisable. Cluster boundaries are soft — customers near boundary regions are genuinely ambiguous and should be treated probabilistically. The absence of demographic data (age, income, region) limits the ability to link behavioural segments to demographic risk factors. Finally, the PCA-2D and 3D projections capture only 56 % and 68 % of total variance respectively; some cluster separation visible in 17-D is invisible in the low-dimensional plots.

Future work includes integrating demographic and time-series data, deploying the fitted model as a REST API to score new customers at onboarding, and using cluster labels as features in a supervised churn or default-prediction model.

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